Jinshi (Peter) Chen

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EDUCATION

Woods Hole Oceanographic Institution, Physical Oceanography — Cambridge & Woods Hole, MA
Massachusetts Institute of Technology, Earth, Atmospheric, and Planetary Sciences — Jun. 2019–Mar. 2025
Doctor of Philosophy
Advisors: Dr. Britt Raubenheimer & Dr. Steve Elgar
Dissertation: *Cross-shore Transformation of Breaking Random Waves in the Surfzone*

Cornell University, Physics — Ithaca, NY
Bachelor of Arts, *magna cum laude* — Aug. 2015–May 2019

POSITIONS

Princeton University, Andlinger Center for Energy and the Environment — Princeton, NJ
Distinguished Postdoctoral Fellow — Apr. 2025–Present

Woods Hole Oceanographic Institution, Applied Ocean Physics & Engineering — Woods Hole, MA
Guest Investigator — Apr. 2025–Present

RESEARCH AREAS & INTERESTS

• Nearshore processes • Air-sea-land interaction • Wave breaking & turbulence • High-fidelity scientific modeling
• Coastal resilience & climate adaptation • Marine renewable energy

PUBLICATIONS

• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2026). Depth-resolved Surfzone Wave and Roller Transformation. *Journal of Geophysical Research: Oceans, 131*(10), e2026JC024082.
• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2025). Anisotropy of Surfzone Turbulence. *Journal of Physical Oceanography, 55*(9), 1435–1450.
• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2024). Wave and Roller Transformation over Barred Bathymetry. *Journal of Geophysical Research: Oceans, 129*(5), e2023JC020413.

MANUSCRIPTS IN PREPARATION

• **Chen, J.**, Scapin, N., & Deike, L. Parameterizing Misaligned Wind-wave Interactions. *In prep.*
• **Chen, J.**, Scapin, N., & Deike, L. Momentum and Energy Exchange in Misaligned Wind-wave Conditions. *In prep.*

INVITED PRESENTATIONS

• *A Peek into Misaligned Oceanic Wind-wave Interactions with High-Fidelity Models,* New Light: Rising Stars in Energy and the Environment, Princeton University, Jun. 2026
• *Cross-shore Transformation of Breaking Random Waves in the Surfzone,* Shanghai Jiao Tong University, Apr. 2025
• *Cross Surfzone Transport Dynamics by Random Breaking Waves,* Sack Lunch Seminar, MIT, Dec. 2024
• *Cross Surfzone Transport Dynamics by Random Breaking Waves,* Center for Coastal Studies Seminar, Scripps Institution of Oceanography, Oct. 2024
• *Simulations and Observations of Surfzone Waves and Undertow*, Coastal Ocean Fluid Dynamics Laboratory (COFDL) Seminar, WHOI, Aug. 2021

CONFERENCE PRESENTATIONS

• **Chen, J.**, Scapin, N., & Deike, L. (2026, May). *A DNS Study of Momentum and Energy Exchange in Misaligned Wind-wave Conditions.* Presented at 2026 International Ocean Vector Winds Science Team Meeting.
• **Chen, J.**, Scapin, N., Wu, J., & Deike, L. (2025, November). *Momentum and Energy Exchange in Misaligned Wind-wave Conditions.* Presented at 2025 Annual Meeting of the APS Division of Fluid Dynamics.
• **Chen, J.**, Raubenheimer, B., Elgar, S., & Tsai, B. (2024, December). *Simulations of Depth-resolved Cross-shore Momentum Transfer in the Surf Zone.* Poster presented at 2024 AGU Fall Meeting.
• **Chen, J.**, Trowbridge, J., Raubenheimer, B., & Elgar, S. (2024, February). *Observations of Surfzone Turbulence Anisotropy*. Presented at 2024 Ocean Sciences Meeting.
• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2023, August). *Tidal Effects of Cross-shore Roller Transformation over Barred Bathymetries*. Presented at 2023 Young Coastal Scientists and Engineers Conference-Americas.
• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2022, December). *Cross-shore Roller Transformation over Barred Bathymetries*. Presented at 2022 AGU Fall Meeting.
• **Chen, J.**, Raubenheimer, B., & Elgar, S. (2018, November). *Surfzone Setup and Alongshore Currents During Hurricane Matthew.* Poster presented at 2018 Annual Meeting of the APS Division of Fluid Dynamics.

RESEARCH GRANTS

• co-PI (through WHOI): Depth-limited waves breaking in viscous nearshore waters, NSF-PO — *In review*
• PI: Ocean Ventures Fund (OVF) award (~$10k) — Jun. 2023
• PI: PADI Foundation Grant (~$8k) — Apr. 2021

HONORS & AWARDS

• MathWorks Fellowship — Aug. 2023–Jun. 2024
• Merrill Presidential Scholar — May 2019
• American Physical Society Division of Fluid Dynamics Student Travel Grant — Sept. 2018
• Woods Hole Oceanographic Institution Summer Student Fellowship — Jun. 2018–Aug. 2018
• China Marine Figure of the Year — Jun. 2014

RESEARCH EXPERIENCE

Princeton University, Andlinger Center for Energy and the Environment — Princeton, NJ
Distinguished Postdoctoral Fellow — Apr. 2025–Present
• Simulate misaligned wind-wave interactions using two-phase DNS with the *Basilisk* solver.
• Develop parameterizations of momentum and energy transfer under misaligned wind-wave conditions.
• Incorporate the modeled results into offshore wind turbine models.

MIT-WHOI Joint Program — Cambridge & Woods Hole, MA
Graduate Student — Dec. 2020–Apr. 2022
• Simulated random vortex generation and advection over a slope using the *Dedalus* framework.
• Derived a dimensionless relationship between vortex properties and slope and bathymetry properties.
• Explored possible critical transitions in vortex advection using a simplified vortex dipole advection model.

MIT-WHOI Joint Program — Cambridge & Woods Hole, MA
Graduate Student — Jun. 2019–Mar. 2025
• Numerically investigated the cross-shore wave roller transformation and vertical structure of undertow with random waves and field-based bathymetry using *OpenFOAM*.
• Investigated the surfzone turbulence structure and anisotropy and their relation to wave and mean current conditions using field observations.

FIELDWORK

SINKEX, Field Research Facility, Duck, NC — Sept. 2023–Oct. 2023
• Deployed vertically aligned ADVs to measure surfzone turbulence (JPO paper, funded by an OVF award).
• Assisted with mounting field cameras to collect surf zone foam images for estimating surface currents.

BIASEX, Field Research Facility, Duck, NC Sept. 2022
• Deployed ADVs and ADCPs in the surf zone.
• Conducted drone operations for remote sensing of surface currents.

DUNEX, Field Research Facility, Duck, NC Aug. 2021–Oct. 2021
• Deployed and maintained ADVs, ADCPs and pressure sensors in the surf zone to measure the vertical structure of undertow (JGR paper, funded by a PADI Foundation award).
• Assisted with mounting field cameras to collect surf zone foam images.

PROFESSIONAL ACTIVITIES

• **Chair & Convener,** Nearshore Processes, 2024 AGU Fall Meeting, Washington, D.C.
• **Reviewer,** *Journal of Atmospheric and Oceanic Technology*

SPECIALIZED SKILLS

Software: MATLAB • Mathematica • FLUENT • LabVIEW
Programming languages: Python • C/C++ • Julia
Numerical models: OpenFOAM • Dedalus • Basilisk
Laboratory: Particle Image Velocimetry (PIV) • Laser-induced Fluorescence (LIF) • Raman spectroscopy
Fieldwork: Acoustic Doppler velocimeters (ADVs) and current profilers (ADCPs) • UAS remote sensing

MENTORING & TEACHING

Guest Instructor, Princeton University Princeton, NJ
• Designed and taught a lecture on theoretical nearshore wave transformation in MAE 330: Ocean Waves. Oct. 2026
• Designed and taught a lecture on nearshore wave and sediment dynamics in MAE 330: Ocean Waves. Oct. 2025

Math Review Instructor, MIT-WHOI Joint Program Woods Hole, MA
• Taught signal processing for incoming MIT-WHOI Joint Program students. Aug. 2024

Graduate Teaching Development Tracks, MIT Cambridge, MA
• Developed teaching skills through interactive learning modules and literature. Nov. 2022–Feb. 2024

Peer Advisor, Cornell University Ithaca, NY
• Mentored six first-year students on selecting physics courses and finding research projects. Aug. 2017–Dec. 2017

Undergraduate Teaching Assistant, Cornell University Ithaca, NY
• Held discussion sessions for PHYS 1112: Physics I: Mechanics and Heat. Feb. 2016–May 2016

AFFILIATIONS

• American Geophysical Union Dec. 2022–Present
• Phi Beta Kappa Honor Society Mar. 2019–Present
• American Physical Society Sept. 2018–Present

ADDITIONAL EXPERIENCE

• **Guest Student,** Woods Hole Oceanographic Institution, Woods Hole, MA Jul. 2016–Aug. 2016
• **Lead Student Researcher on Cyanobacteria,** Tsinghua University, Beijing, China Sept. 2013–Aug. 2014

CERTIFICATIONS

• **MIT Grant Writing Certificate** Jun. 2024
• **MIT Research Mentoring Certificate** Apr. 2024
• **MIT Graduate Teaching Certificate** Apr. 2024
• **Scientific Diver,** American Academy of Underwater Sciences Jul. 2021